CASE STUDY 03

CISCO PACKET TRACER

DNS SERVER CONFIGURATION

& NAME RESOLUTION

Hands-on lab evidence: static IPv4 addressing, DNS A record, client resolver settings, and validated name-to-address connectivity.

VALIDATED RESULT

www.test.com -> 192.168.1.10

DNS lookup succeeded | 4/4 ICMP replies | 0% packet loss



SUMMER BAHAM

IT Support Engineering & Cybersecurity Portfolio

SIMULATED NETWORK ENVIRONMENT / EVIDENCE-BASED LAB

01 / LAB ENVIRONMENT

Topology & addressing

One DNS server and one client connected through a Cisco 2960-24TT switch on a single /24 LAN.

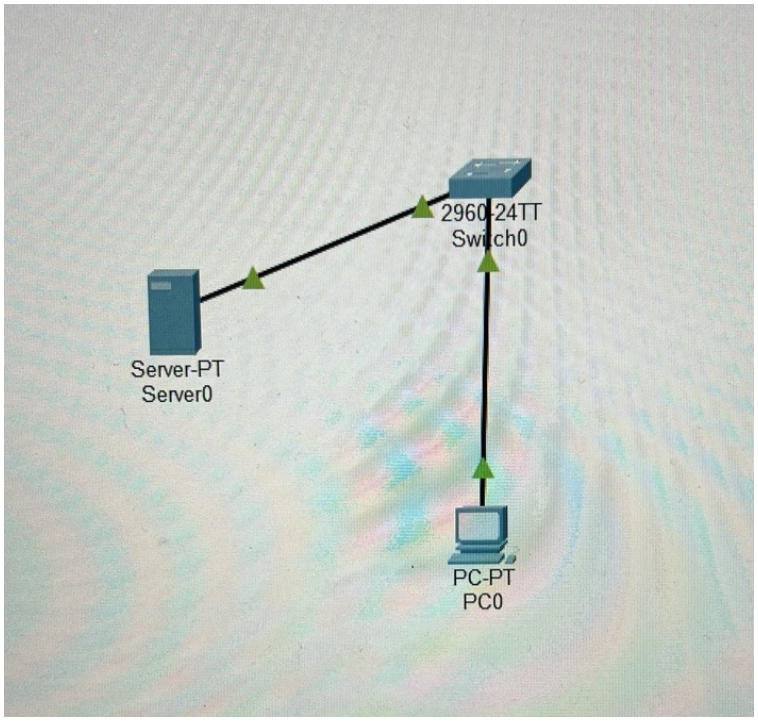


Figure 1. Cisco Packet Tracer topology: Server0 - Switch0 - PC0

NETWORK ADDRESSING

DEVICE	IPv4 ADDRESS	SUBNET MASK	DNS SERVER
Server0	192.168.1.10	255.255.255.0	192.168.1.10
PC0	192.168.1.20	255.255.255.0	192.168.1.10

DESIGN NOTE

No router is present in this topology. Server0 and PC0 are on the same subnet, so no default gateway is required for their local communication.

Server0 static IPv4 settings

Configure the DNS host with a stable IPv4 address before exposing the DNS service to the client.

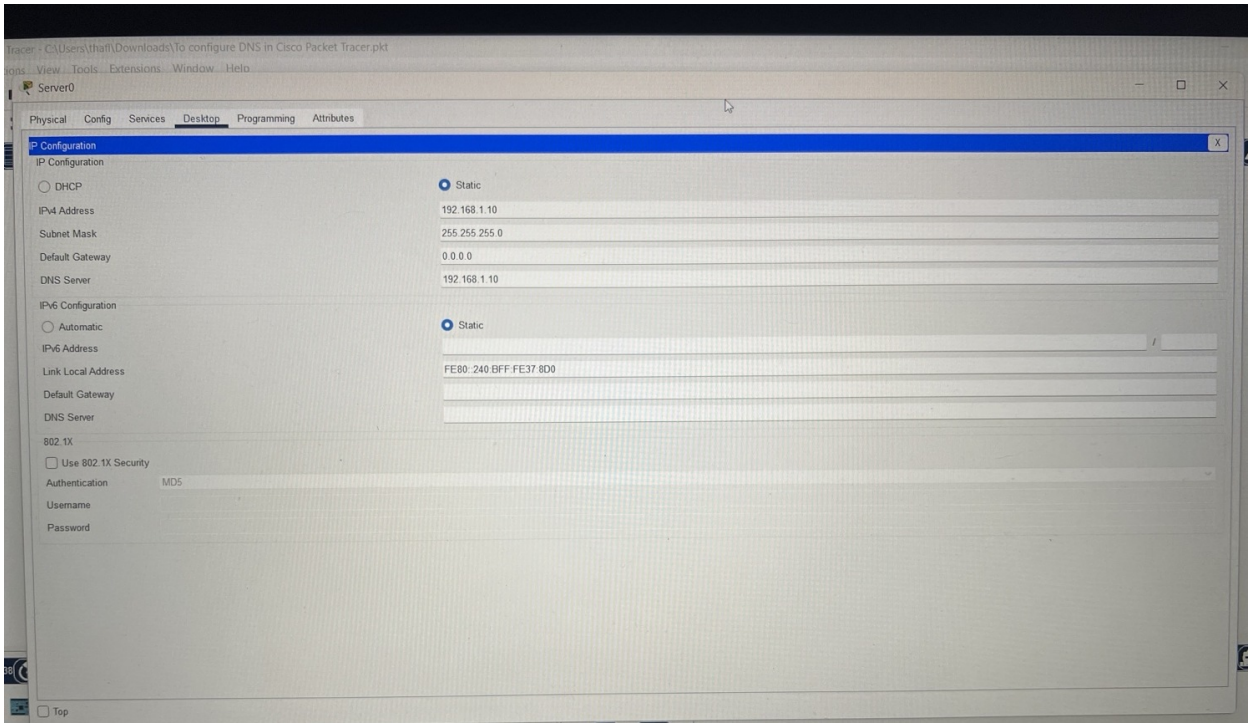


Figure 2. Server0 > Desktop > IP Configuration

OBSERVED CONFIGURATION

SETTING	CONFIGURED VALUE
Address assignment	Static
IPv4 address / prefix	192.168.1.10 / 24
DNS server field	192.168.1.10 (self)

Default gateway: 0.0.0.0 (not used on this single-subnet LAN)

DNS service & A record

DNS is enabled on Server0. The configured A record maps a host name to the server IPv4 address.

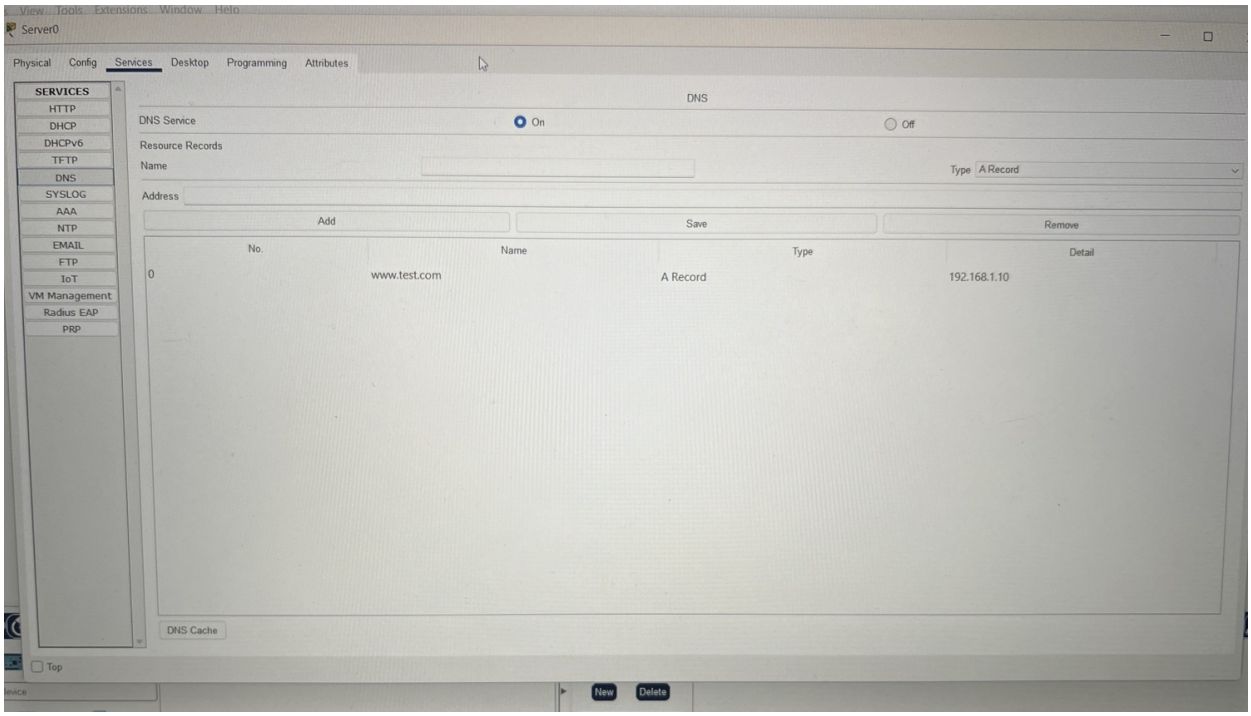


Figure 3. Server0 > Services > DNS (enabled, A record displayed)

VERIFIED RESOURCE RECORD

www.test.com → **192.168.1.10**

Type: A record

DNS service: On

WHAT THIS TESTS

An A record provides an IPv4 destination for a hostname. Client success depends on both the server-side record and the client pointing to the correct DNS server.

PC0 settings & DNS lookup

PC0 uses Server0 as its DNS resolver, and the command-line query returned the intended A-record address.

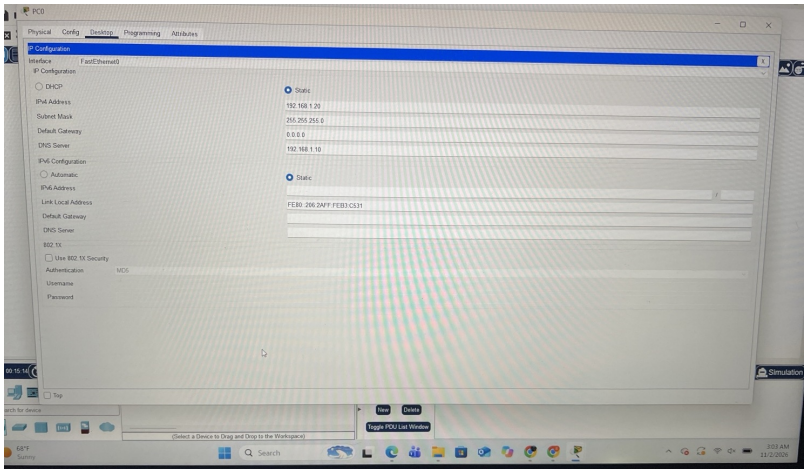


Figure 4. PC0 > Desktop > IP Configuration

CLIENT	192.168.1.20/24	**DNS RESOLVER**	192.168.1.10

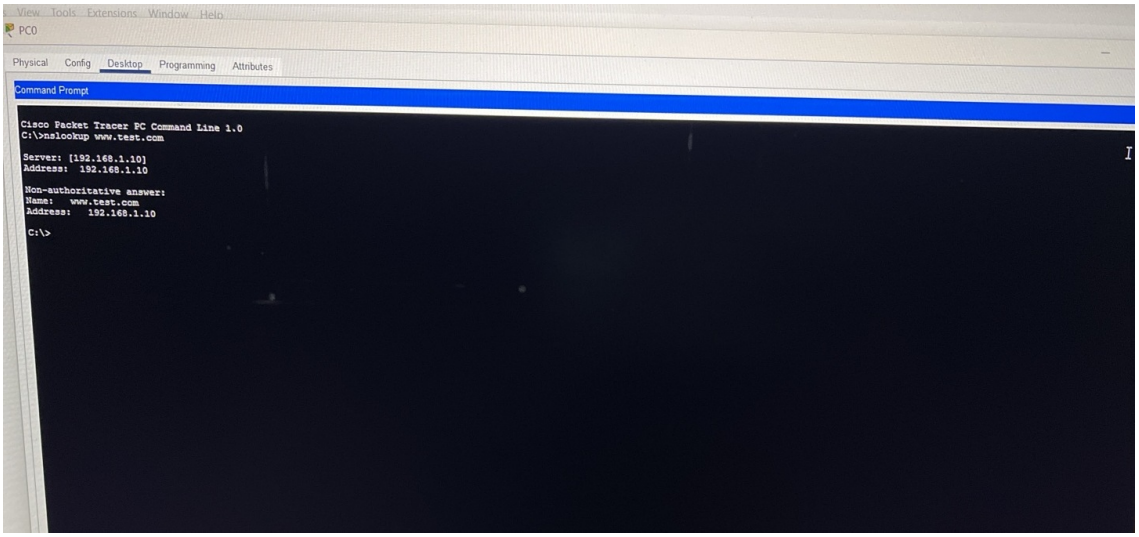


Figure 5. nslookup www.test.com returned 192.168.1.10

INTERPRETATION

The queried name resolved to 192.168.1.10 via 192.168.1.10, confirming DNS lookup success from PC0.

Connectivity verified

A final ICMP test used the hostname rather than the raw server address, checking both resolution and reachability.

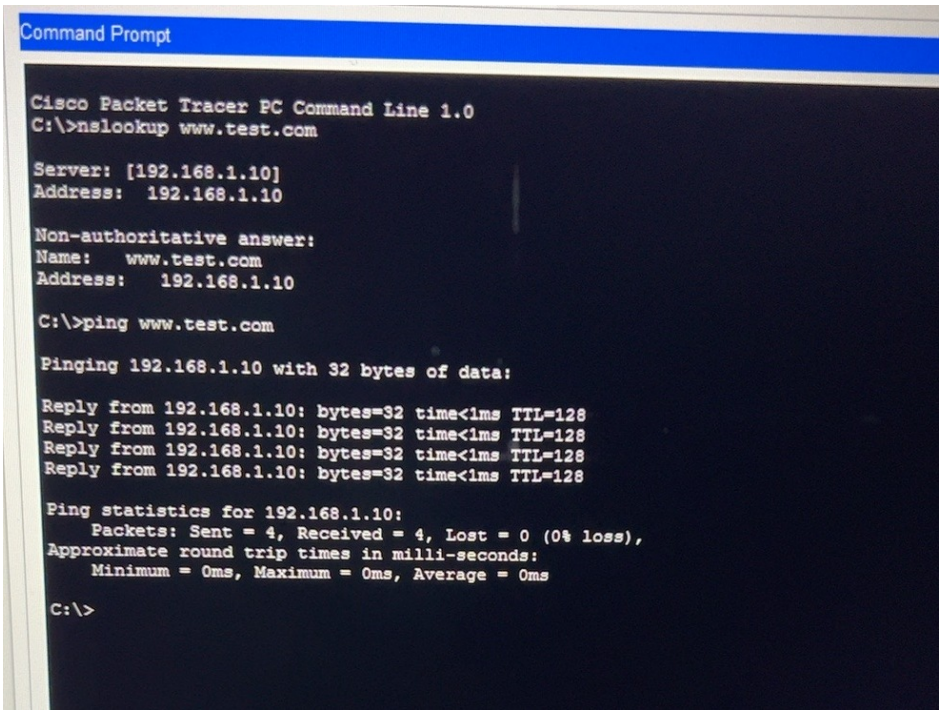


Figure 6. PC0 command prompt - nslookup and ping results

TEST OUTCOME

4 / 4 replies received

0% packet loss

EVIDENCE SUMMARY

- **DNS query: PASS**
  PC0 resolved www.test.com to 192.168.1.10.
- **Host-based ping: PASS**
  Four packets sent and received; 0% loss to 192.168.1.10.

Scope: This lab verifies a local, simulated DNS setup. It does not demonstrate a public DNS deployment, internet name resolution, or production availability.